Problem 2. Rectangle Tree

Input file:	*standard input*
Output file:	*standard output*
Time limit:	6 seconds
Memory limit:	512 mebibytes

Mr. Peanutbutter has recently discovered a nice $n \times n$ field covered with various crops. Diane told Mr. Peanutbutter that for generations this particular field is planted with crops using tree-like rectangle method. While Mr. Peanutbutter got distracted by a bird, Diane continued.

A *combinatorial rectangle* in the field is a subset of squares of the field of the form $A \times B$ where A and B are subsets of the set $\{0, \ldots, n-1\}$.

A *rectangle tree* is a rooted binary tree with k vertices with the following properties. Each vertex v of the tree is labeled with a combinatorial rectangle $r(v) \subseteq \{0, \ldots, n-1\} \times \{0, \ldots, n-1\}$. If s is an inner node of the tree, and c_1 and c_2 are its direct descendants, then their combinatorial rectangles form a partition of $r(s)$: formally, $r(s) = r(c_1) \cup r(c_2)$ and $r(c_1) \cap r(c_2) = \varnothing$. A node cannot have only one direct descendant.

Let $\mathrm{Crop}(x, y)$ be the crop that grows on the square $(x, y) \in \{0, \ldots, n-1\} \times \{0, \ldots, n-1\}$ of the field. A rectangle tree T with the root Root *computes* the crop types of the field if $r(\mathrm{Root}) = \{0, \ldots, n-1\} \times \{0, \ldots, n-1\}$ and for each leaf ℓ, the combinatorial rectangle $r(\ell)$ has exactly one type of crop growing on it: that is, for any two $(x, y), (x', y') \in r(\ell)$, we have $\mathrm{Crop}(x, y) = \mathrm{Crop}(x', y')$.

The *depth* of tree T is the largest distance between the root of T and a leaf of T. Here, distance stands for the number of edges in the shortest path between the vertices.

The *size* of tree T is the number of vertices in it.

You are given a rectangle tree T computing crop types Crop. Let the size of T be S. Construct another rectangle tree T' computing Crop such that its depth is at most $3 \log_2 S$ and its size is at most $5S$.

Input

The first line contains a single integer n, the size of the field ($1 \le n \le 1000$). Each of the next n lines contain n integers describing the types of the crops. The j-th integer in the i-th row is the type of crop in the square (i, j). All types are positive integers not exceeding 10^7.

The next line contains a single integer S, the size of the rectangle tree ($1 \le S \le 10\,000$, $S \cdot n \le 10^6$). The i-th of the next S lines contains the description of the i-th vertex of the tree. It contains several space-separated integers: p, m_1, m_2, a_1, ..., a_{m_1}, b_1, ..., b_{m_2}. Here, $p \in \{0, 1, \ldots, S-1\}$ is the number of the parent of vertex i (if i is the root, then $p = i$), and the combinatorial rectangle corresponding to this vertex is $r(i) = \{a_1, \ldots, a_{m_1}\} \times \{b_1, \ldots, b_{m_2}\}$.

It is guaranteed that, if ℓ is a leaf of the tree, then all types of crops in $r(\ell)$ are the same. Additionally, for each inner vertex v with direct descendants c_1 and c_2, the rectangles $r(c_1)$ and $r(c_2)$ form a partition of $r(v)$: $r(v) = r(c_1) \cup r(c_2)$ and $r(c_1) \cap r(c_2) = \varnothing$. Finally, if i is the root, then $r(i) = \{0, \ldots, n-1\} \times \{0, \ldots, n-1\}$.

Output

Print a tree of depth at most $3 \log_2 S$ and of size at most $5S$ such that it is also a rectangle tree that computes the given crops. The tree should be printed in the same format as the one given in the input.

Example

standard input	standard output
3 1 1 2 1 1 2 2 2 2 5 0 3 3 0 1 2 0 1 2 0 3 2 0 1 2 0 1 0 3 1 0 1 2 2 1 2 2 0 1 0 1 1 1 2 2 0 1	7 0 3 3 0 1 2 0 1 2 0 1 3 2 0 1 2 1 1 2 2 0 1 1 1 1 2 2 0 2 3 0 1 0 1 2 4 2 1 0 1 2 4 2 2 0 1 0 1

Problem 3. Closest Pair Algorithm

Input file:	*standard input*
Output file:	*standard output*
Time limit:	10 seconds
Memory limit:	512 mebibytes

There is a classic problem about finding two closest points amongst a set of points on a plane. There is also a classic randomized *algorithm* to solve it, which goes like this:

```
rotate the plane around the origin by a random angle phi
let p be the array of points
sort p by x coordinate in increasing order
ans = INF;
for (int i = 0; i < n; i++) {
    for (int j = i - 1; j >= 0; j--) {
        if (p[i].x - p[j].x >= ans) break;
        ans = min(ans, dist(p[i], p[j]));
    }
}
```

Here "`INF`" is some number greater than all distances. The function "`dist`" returns Euclidean distance between the given points. Note that all x coordinates are distinct after rotation with probability 1.

I know that the algorithm works well in practice, but how well exactly? I ask you to compute the expected number of calls of the function "`dist`", assuming that the angle "`phi`" is chosen uniformly at random from the range $[0; 2 \cdot \pi)$.

Input

The first line contains a single integer n ($2 \le n \le 250$) — the number of points.

The next n lines contains coordinates of points, one per line.

All the coordinates are not greater than 10^6 by absolute value.

It is guaranteed that all points are distinct. It is guaranteed that no three points lie on the same line.

Output

Print one number — the expected number of calls of the function "`dist`".

Your answer is considered correct if its absolute or relative error does not exceed 10^{-6}.

Formally, let your answer be a, and the jury's answer be b. Your answer is accepted if and only if $\frac{|a-b|}{\max(1,|b|)} \le 10^{-6}$.

Examples

standard input	standard output
4 0 0 0 1 1 0 1 1	5.0000000000000
3 0 0 0 1 1 0	2.5000000000000

Problem 5. Fast as Ryser

Input file:	*standard input*
Output file:	*standard output*
Time limit:	4 seconds
Memory limit:	512 mebibytes

After reading the paper *Counting Perfect Matchings as Fast as Ryser*, you learned how to count the number of perfect matchings in a general graph in $O(2^n n^2)$. So you decided to write this problem to encourage people to read the paper and learn new technology.

You are given an undirected graph with n vertices and m edges, and also a constant c. We define the weight of an edge set S as follows:

- If there are two edges in set S sharing common vertices, the weight is 0.
- Otherwise, the weight is $c^{|S|}$. Note that the weight of an empty set is 1.

Compute the sum of the weight of all subsets of edges. The answer can be large, so output it modulo $10^9 + 7$.

Input

The first line contains three integers n, m, c ($1 \le n \le 36$, $0 \le m \le \frac{n(n-1)}{2}$, $1 \le c \le 10^9 + 6$).

Each line of the following m lines contains two integers u, v ($1 \le u, v \le n$, $u \ne v$) indicating an undirected edge (u, v) in the graph. All edges are distinct.

Output

Output one integer: the answer.

Examples

standard input	standard output
6 10 100 3 6 1 3 2 4 3 4 4 6 1 2 4 5 2 3 1 4 3 5	2171001
8 11 818466928 6 7 3 6 6 5 7 3 6 2 8 1 1 7 4 3 5 1 6 1 6 4	425176360

Problem 7. Interesting Game

Input file:	*standard input*
Output file:	*standard output*
Time limit:	2 seconds
Memory limit:	512 mebibytes

50th IMO 2009 — *Combinatorics* — **C5**

C5 NLD (Netherlands)

Five identical empty buckets of 2-liter capacity stand at the vertices of a regular pentagon. Cinderella and her wicked Stepmother go through a sequence of rounds: At the beginning of every round, the Stepmother takes one liter of water from the nearby river and distributes it arbitrarily over the five buckets. Then Cinderella chooses a pair of neighboring buckets, empties them into the river, and puts them back. Then the next round begins. The Stepmother's goal is to make one of these buckets overflow. Cinderella's goal is to prevent this. Can the wicked Stepmother enforce a bucket overflow?

Is it a notorious coincidence with this problem?

Cinderella and her wicked Stepmother are playing the game. Cinderella has n non-negative integers $a_1, a_2, \ldots, a_n$ at first. There are two parameters A and B for this game.

Cinderella and Stepmother take turns playing, starting with Cinderella. One each turn, Cinderella can replace the sequence $a_1, a_2, \ldots, a_n$ by a new **integer** sequence $a'_1, a'_2, \ldots, a'_n$ such that

- $a'_1 \geq a_1, \ldots, a'_n \geq a_n$
- $\sum_{i=1}^{n} a'_i \leq \sum_{i=1}^{n} a_i + A$

Then Stepmother can choose B indices $i_1, i_2, \ldots, i_B$, and set $a_{i_1}, a_{i_2}, \ldots, a_{i_B}$ to 0.

The game continues forever. Let M be the maximum value of $a_1, a_2, \ldots, a_n$ for all the time. Cinderella wants to maximize M, and Stepmother wants to minimize M.

Determine the value of M if both players play optimally.

Input

The first line contains an integer T ($1 \leq T \leq 10^5$) indicating the number of test cases. For each test case:

The first line contains three integers n, A, B ($1 \leq B \leq n \leq 10^5$, $0 \leq A \leq 10^{12}$).

The second line contains n integers $a_1, a_2, \ldots, a_n$ ($0 \leq a_i \leq 10^{12}$).

It is guaranteed that $\sum n \leq 5 \times 10^5$.

Output

For each test case, output a line containing one integer: the answer.

Example

standard input	standard output
4 3 5 1 1 2 3 5 5 1 0 2 1 0 3 5 100 5 1 2 3 4 5 8 3 1 5 1 2 2 0 2 5 1	11 14 105 9

Note

A possible game process for the first test case:

$\{1, 2, 3\} \to \{3, 4, 4\} \to \{3, 4, 0\} \to \{6, 6, 0\} \to \{6, 0, 0\} \to \{11, 0, 0\}$.

Problem 11. Knowledge-Oriented Problem

Input file:	*standard input*
Output file:	*standard output*
Time limit:	5 seconds
Memory limit:	512 mebibytes

After gaining a lot of knowledge, you decided to write a knowledge-oriented problem.

You are given an undirected graph G with n vertices and m edges. You copy it k times and denote the copies by $G_1, G_2, \ldots, G_k$. You add edges between vertex u in copy G_i and the same vertex u in copy G_{i+1} for all $1 \le i \le k-1$ and $1 \le u \le n$.

Find the number of spanning trees of the new graph. The answer can be large, so output it modulo $10^9 + 7$.

Input

The first line contains three integers n, m, k ($1 \le n \le 500$, $0 \le m \le \frac{n(n-1)}{2}$, $1 \le k \le 10^{18}$).

Each of the following m lines contains two integers u, v ($1 \le u, v \le n$, $u \ne v$) indicating an undirected edge (u, v) in the graph. All edges are distinct.

Output

Output one integer: the answer.

Examples

standard input	standard output
5 6 2 3 2 5 1 3 4 2 4 5 3 1 3	4725
2 1 200 1 2	272581704
5 10 1000000000000000000 1 2 1 3 1 4 1 5 2 3 2 4 2 5 3 4 3 5 4 5	569698435

Problem 13. LCM Sum

Input file:	*standard input*
Output file:	*standard output*
Time limit:	7 seconds
Memory limit:	512 mebibytes

Are you sick of solving problems like computing the prefix sum of a random number theory function? As a terrible problem writer, here I present another one for you.

Compute

$$\sum_{x=1}^{n} \operatorname{lcm}(x, x+1, \ldots, x+k).$$

The answer can be large, so output it modulo $10^9 + 7$.

Input

The first line contains two integers n, k ($1 \le n \le 10^{18}$, $0 \le k \le 30$).

Output

Output one integer: the answer.

Examples

standard input	standard output
10 3	18936
10000 6	43482752
1000000000 15	688102997

Problem 17. Evacuation

Input file:	*standard input*
Output file:	*standard output*
Time limit:	6 seconds
Memory limit:	1024 mebibytes

There are $N + 2$ towns in a straight line. The towns are numbered from 0 through $N + 1$ from left to right. Each town i ($1 \leq i \leq N$) has a shelter which can contain up to A_i people.

Currently, S people are traveling together and visiting one of the towns. However, you don't know which town they are visiting.

You just got to know that there are Q meteorites that can hit the towns. The i-th meteorite may strike towns $L_i, L_i + 1, \cdots, R_i$. To ensure the safety of the travelers, for each meteorite, you want to calculate the **evacuation cost**.

The evacuation cost for a meteorite is calculated as follows:

- The evacuation cost is the minimum total cost needed to make all travelers **safe** no matter which town they are visiting.
- A person is safe if he/she is in a shelter or a town outside the effect of the meteorite.
- It takes 1 unit cost to move one person to an adjacent town (two towns are adjacent iff their numbers differ by exactly 1).

Note that only moving people costs money, and other actions (like accommodating people in a shelter) don't. It is guaranteed that towns 0 and $N+1$ will never be affected by meteorites, so it is always possible to make all travelers safe.

Write a program that, for each meteorite, calculates the evacuation cost for that meteorite.

Input

Input is given from Standard Input in the following format:

$N\ S$

$A_1\ A_2\ \cdots\ A_N$

Q

$L_1\ R_1$

$L_2\ R_2$

$\vdots$

$L_Q\ R_Q$

Constraints:

- $1 \leq N \leq 2 \times 10^5$
- $1 \leq S \leq 10^{12}$
- $0 \leq A_i \leq S$
- $A_1 + A_2 + \cdots + A_N \leq 10^{12}$
- $1 \leq Q \leq 2 \times 10^5$
- $1 \leq L_i \leq R_i \leq N$

- All values in input are integers.

Output

Print Q lines. The i-th line should contain the evacuation cost for the meteorite i.

Example

standard input	standard output
5 10 3 1 1 4 1 3 1 4 3 5 2 5	14 10 13

Note

- For the first meteorite, it takes 14 unit costs when the travelers are visiting the town 2.
- For the second meteorite, it takes 10 unit costs when the travelers are visiting the town 4.
- For the third meteorite, it takes 13 unit costs when the travelers are visiting the town 3.

Problem 19. Matrix Inversion

Input file:	*standard input*
Output file:	*standard output*
Time limit:	1 second
Memory limit:	1024 mebibytes

You have an N by N grid board, which is initially empty. You will write an integer to each cell, using each integer from 1 to N^2 exactly once. Let $M_{i,j}$ be the integer written on the cell in the i-th row from the top and the j-th column from the left.

Let's define sequences A and B as follows:

- $A = M_{1,1}, M_{1,2}, \ldots, M_{1,N}, M_{2,1}, M_{2,2}, \ldots, M_{2,N}, \ldots, M_{N,N}$
- $B = M_{1,1}, M_{2,1}, \ldots, M_{N,1}, M_{1,2}, M_{2,2}, \ldots, M_{N,2}, \ldots, M_{N,N}$

For example, when the board looks like this,

```
1 3 4
2 7 6
9 8 5
```

A and B are defined as follows.

- $A = 1, 3, 4, 2, 7, 6, 9, 8, 5$
- $B = 1, 2, 9, 3, 7, 8, 4, 6, 5$

You are given integers N, X, Y. Find a way to fill in the cells so that the inversion numbers of A and B are X and Y respectively, or report that it is impossible to do so.

Note: an inversion number of a sequence $C = c_1, c_2, \ldots, c_{N \times N}$ is the number of pairs (i, j) s.t. both $i < j$ and $c_i > c_j$ are satisfied.

Input

Input is given from Standard Input in the following format:

N X Y

Constraints:

- $2 \leq N \leq 300$
- $0 \leq X, Y \leq \frac{N^2(N^2-1)}{2}$

Output

If there is no solution, print '`No`'.

Otherwise, print the answer in the following format:

Yes

$M_{1,1}$ $M_{1,2}$ $\ldots$ $M_{1,N}$

$M_{2,1}$ $M_{2,2}$ $\ldots$ $M_{2,N}$

$\vdots$

$M_{N,1}$ $M_{N,2}$ $\ldots$ $M_{N,N}$

If there are multiple solutions, you can print any of them.

Examples

standard input	standard output
3 8 13	Yes 1 3 4 2 7 6 9 8 5

Problem 23. Game and Queries

Input file:	*standard input*
Output file:	*standard output*
Time limit:	2 seconds
Memory limit:	1024 mebibytes

Alice and Bob are going to play a game. The rule of the game is as follows:

- There are some monsters, each of which has a specified HP.
- Alice and Bob take turns alternately, with Alice going first.
- In Alice's turn, she chooses one of the monsters and increases its HP by 1.
- In Bob's turn, he chooses one of the monsters and decreases its HP by 2. If the monster's HP becomes 0 or less, the monster disappears.
- The game ends when k monsters disappear.

Alice's objective is to finish the game as late as possible, while Bob's is as soon as possible.

Initially, there are no monsters. You have to process Q queries of the following types:

- Type 1: You are given two integers X_i and Y_i. After this query, the number of monsters whose HPs are X_i becomes Y_i.
- Type 2: Given an integer K_i, calculate how many turns would Bob take if the game started with current monsters and $k = K_i$, assuming both players play optimally.

Note that the game doesn't happen in reality, and the monsters don't disappear.

Input

Input is given from Standard Input in the following format:

Q

Description of the 1-st query

Description of the 2-nd query

$\vdots$

Description of the Q-th query

The description of each query is in one of the following formats:

Type 1: 1 X_i Y_i

Type 2: 2 K_i

Constraints:

- $1 \leq Q \leq 3 \times 10^5$
- $1 \leq T_i \leq 2$
- $1 \leq X_i \leq 10^6$
- $0 \leq Y_i \leq 10^6$
- $1 \leq K_i \leq$ (the number of current monsters)

- There is at least one query of the type 2
- All values in input are integers.

Output

For each query of the type 2, print the answer in a line.

Example

standard input	standard output
6 1 1 4 2 3 1 2 3 2 6 1 2 2 2 6	3 7 8
20 1 1 12 2 12 1 2 15 2 12 2 3 1 12 10 2 27 1 14 6 2 7 2 43 2 22 1 8 7 1 1 11 2 49 1 5 19 2 38 2 8 1 12 14 1 16 1 2 24	12 12 3 42 7 246 25 301 91 8 32

Note

After the 5-th query, there are 4 monsters whose HPs are 1 and 2 monsters whose HPs are 2.

Problem 29. Deja Vu

Input file:	*standard input*
Output file:	*standard output*
Time limit:	5 seconds
Memory limit:	512 mebibytes

You are given an array $x_1, x_2, \ldots, x_n$.

You need to perform two types of queries on this array.

- Given i and y, set $x_i = y$.
- Given l, find the smallest d among all tuples (a, b, c, d) with $l \le a < b < c < d$ and $x_a < x_b < x_c < x_d$, or reply that there are no such tuples.

Input

The first line contains two integers n, q ($1 \le n, q \le 500\,000$): the number of elements in the array and the number of queries.

The second line contains n integers $x_1, x_2, \ldots, x_n$ ($1 \le x_i \le 10^9$).

Each of the next q lines contains the description of a query.

If the first integer in the line is equal to 1, then the next two integers are i and y ($1 \le i \le n$, $1 \le y \le 10^9$), describing a query of the first type.

Otherwise, the first integer in the line is equal to 2, and the next integer is equal to l ($1 \le l \le n$), describing a query of the second type.

Output

For each query of the second type, return the smallest d among all tuples (a, b, c, d) such that $l \le a < b < c < d$ and $x_a < x_b < x_c < x_d$, or print "-1" if there are no such tuples.

Example

standard input	standard output
11 10 1 2 3 4 5 10 9 8 7 6 8 2 1 1 3 2 2 1 1 1 2 2 1 2 5 2 6 1 9 6 1 10 7 2 5	4 5 6 -1 -1 11

Problem 31. Island

Input file:	*standard input*
Output file:	*standard output*
Time limit:	7 seconds
Memory limit:	512 mebibytes

Welcome to island Bitcairn! We have everything here – settlements, roads, beautiful lake, the Internet and a monster living in the lake which is ready to destroy the whole island! Wait, wait, wait – a monster in the lake?!

Byteasar, governor of Bitcairn, has just ordered you to prepare an evacuation plan for tourists in a case of monster's attack. He told you following information about the island:

- The island can be imagined as a ring whose outer boundary is the boundary of a sea and inner boundary is a boundary of the lake.
- There are n settlements numbered with consecutive numbers from 1 to n.
- Settlements with numbers from 1 to a are placed on the boundary of the lake. Moreover they are numbered either clockwise or counterclockwise along that boundary.
- Settlements with numbers from $a + 1$ to $a + b$ are placed on a coast (boundary of sea). Moreover they are numbered either clockwise or counterclockwise along the coast.
- There are m roads connecting settlements. Each of the roads can't pass through neither the lake, the sea nor other settlements. Moreover two roads can intersect in their common endpoint only (if they share one). In other words road network forms a planar graph. Moreover each road can be either directed or undirected.
- All tourists live in settlements that are placed on the boundary of the lake. It can be assumed that from each of these settlements it is possible to get to the coast using road network (possibly using multiple roads).

In order to enable evacuation, you need to design a plan of building seaports. Such plan should contain subset of coastal settlements where seaports will be built and it ensures safety for all tourists if and only if each tourist living in settlement on the boundary of the lake can reach at least settlements on the boundary of the sea where seaports will be built.

Byteasar would like to know, what is the number of plans ensuring safety? Since the result can be huge, it is sufficient to print it modulo $10^9 + 7$. You need to hurry up – safety of tourists depends on you!

Input

The first line of input contains four intergers n, m, a and b ($2 \leq n \leq 500\,000$, $1 \leq m \leq 1\,000\,000$, $a, b \geq 1$, $a + b \leq n$) denoting number of settlements, number of roads, number of settlements on the boundary of the lake and number of settlements on the boundary of the sea, respectively.

Following m lines describe roads. Each of them contains a description of one road in the following format:

- u_i `--` v_i (describing bidirectional road connecting settlements u_i and v_i)
- u_i `->` v_i (describing unidirectional road going from settlement u_i to settlement v_i)

where $1 \leq u_i, v_i \leq n$ and $u_i \neq v_i$.

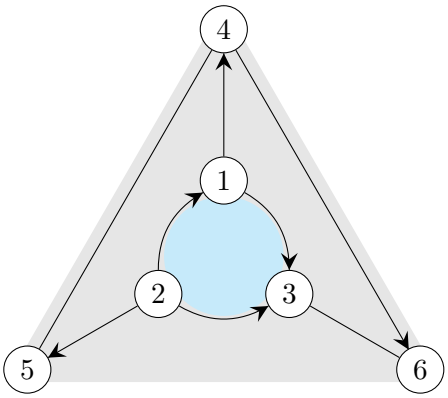
No two roads connect the same pair of settlements. You can assume that these settlements and roads form a planar graph. Moreover for every settlement on the boundary of the lake at least one settlement on the coast can be reached.

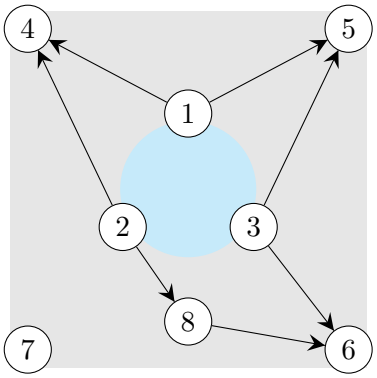
Output

You need to output one integer – number of plans ensuring safety, modulo $10^9 + 7$.

Examples

standard input	standard output
6 8 3 3 2 -> 1 2 -> 3 1 -> 3 3 - 6 1 -> 4 2 -> 5 4 -> 6 4 - 5	4
8 7 3 4 1 -> 4 1 -> 5 2 -> 4 2 -> 8 3 -> 6 3 -> 5 8 -> 6	8





Explanation to sample tests: In the first test we need to build a seaport in a settlement number 6 in order to ensure safety for tourists living in settlement number 3. However, tourists from settlements number 1 and 2 are able to reach it as well, so they safety will be ensured as well in that case, so it doesn't matter whether we build seaports in settlements number 4 and 5.

In the second sample test we need to build seaports in at least two out of settlements in numbers 4, 5 and 6 and that is sufficient to ensure safety of all tourists. It doesn't matter whether we build seaport in settlement number 7. It can be readily checked that in total there are 8 plans ensuring safety.

Problem 37. Easy One

Input file:	*standard input*
Output file:	*standard output*
Time limit:	3 seconds
Memory limit:	512 mebibytes

You have a sequence of digits 1 and 2. In one step you can:

1. Insert 1 in a place which is to the right of every other 1 (or anywhere if there are no 1s).
2. Transform any 2 into 1, if there are no 1s to the right of this 2.
3. Delete the rightmost 1 (note that this operation is inverse to the operation 1).
4. Transform the rightmost 1 into 2 (note that this operation is inverse to the operation 2).

For example, you can obtain the following sequences in one step from the sequence 11212122:

- With operation 1: 112121**1**22, 112121212, 112121222**1**.
- With operation 2: 11212112, 1121212**1**.
- With operation 3: 1121222.
- With operation 4: 11212**2**22.

Your task is to calculate the number of ways to transform a sequence of exactly a digits 2 to a sequence of exactly b digits 2, using exactly t operations.

Input

The only line of the input contains three integers a, b, and t ($0 \le a, b, t \le 10^6$).

Output

Output the number of ways to obtain a sequence of b digits 2 from a sequence of a digits 2 in exactly t steps. As this number can be very large, output it modulo prime number 998 244 353.

Examples

standard input	standard output
0 0 4	3
1 4 6	60

Note

In the first sample you should obtain an empty sequence from an empty sequence in 4 steps. Ways to do this are (ε stands for empty sequence):

$$\varepsilon \to 1 \to \varepsilon \to 1 \to \varepsilon$$
$$\varepsilon \to 1 \to 11 \to 1 \to \varepsilon$$
$$\varepsilon \to 1 \to 2 \to 1 \to \varepsilon$$

Problem 41. Hardcore String Counting 2

Input file:	*standard input*
Output file:	*standard output*
Time limit:	1 second
Memory limit:	512 mebibytes

What happened to "Hardcore String Counting 1"? That's a secret!

A non-empty word v over some alphabet is a *square* if it can be represented as $v = ww$ for some word w.

A word is *square-free* if all its non-empty substrings are not squares.

Your task is to compute the number of square-free words of length ℓ over the alphabet $\{\mathtt{a}, \mathtt{b}, \mathtt{c}\}$ for each ℓ from 1 to n.

Input

The only line of the input contains an integer n ($1 \le n \le 120$).

Output

For each ℓ from 1 to n, print the number of square-free words of length ℓ over the alphabet $\{\mathtt{a}, \mathtt{b}, \mathtt{c}\}$ on a separate line.

Example

standard input	standard output
10	3 6 12 18 30 42 60 78 108 144

Note

You can see `https://oeis.org/A006156` for more details, but this probably won't help you much.

Problem 43. The Missing Pet

Input file:	*standard input*
Output file:	*standard output*
Time limit:	4 seconds
Memory limit:	512 mebibytes

You are given an $n \times n$ chessboard, the rows and columns of which are numbered from 1 to n respectively. Little Q punched several holes in specified locations: the i-th hole is located at (r_i, c_i).

Little Q also has a pet dog. Now, the dog is getting lost on the chessboard at the cell (r_0, c_0). It will move to a random adjacent cell every second. Each of the adjacent cells is selected with equal probability. Here, two cells are adjacent if they share a common edge. If the dog arrives at a cell with a punched hole, it will fall into the hole.

Now, Little Q is wondering: for each hole, what is the expected number of seconds that the pet walks on the chessboard, given that it finally falls into this hole? Please help him.

Input

The first line contains an integer T ($1 \le T \le 20$), the number of test cases. For each test case:

The first line contains two integers n and k ($2 \le n \le 200$, $1 \le k \le 200$) indicating the size of the given chessboard and the number of holes.

Then k lines follow, the i-th of which contains two integers r_i and c_i ($1 \le r_i, c_i \le n$) indicating the location of the i-th hole.

The last line of each test case contains two integers r_0 and c_0 ($1 \le r_0, c_0 \le n$) denoting the starting location of the pet.

It is guaranteed that all given holes are distinct, and the pet is not located at a hole initially. It is also guaranteed that $\max(n, k) > 5$ holds in at most one test case.

Output

For each test case, output a single line with k integers: for each hole, in the order they are given in the input, print the expected number of seconds the pet walks on the chessboard, given that it finally falls into this hole.

More precisely, if a hole is reachable and the reduced fraction of the expected number of seconds is $\frac{p}{q}$, you should output the minimum non-negative integer r such that $q \cdot r \equiv p \pmod{10^9 + 7}$. You may safely assume that such r always exists in all test cases. If a hole is unreachable, output "`-1`" instead.

Example

standard input	standard output
2 3 3 1 1 1 2 2 1 2 2 5 3 5 3 4 1 3 2 4 5	-1 4 4 669185882 381533358 341349117

Problem 47. Fibonacci Partition

Input file:	*standard input*
Output file:	*standard output*
Time limit:	5 seconds
Memory limit:	256 mebibytes

The sequence of Fibonacci numbers is defined as:

$$F_n = \begin{cases} 1 & n = 1 \\ 2 & n = 2 \\ F_{n-1} + F_{n-2} & \text{otherwise} \end{cases}$$

The first few elements of the sequence are $1, 2, 3, 5, 8, 13, 21, 34, \ldots$

For a given positive integer n, let $partition(n)$ be the maximum value of m such that n can be expressed as a sum of m distinct Fibonacci numbers. For example, $partition(1) = partition(2) = 1$, $partition(3) = partition(4) = partition(5) = partition(7) = 2$, $partition(6) = partition(8) = 3$.

Chiaki has an integer X which initially equals to 0. She will perform some operations on X: the i-th operation will add $a_i \cdot F_{b_i}$ to X.

After each operation, Chiaki would like to know the value of $partition(X)$. It is guaranteed that, after each operation, X will be a positive integer.

Input

There are multiple test cases. The first line of input contains an integer T, indicating the number of test cases. For each test case:

The first line contains an integer n ($1 \le n \le 5 \cdot 10^4$): the number of operations.

Each of the next n lines contains two integers a_i and b_i ($1 \le |a_i|, b_i \le 10^9$).

It is guaranteed that the sum of n for all test cases will not exceed $5 \cdot 10^4$.

Output

For each test case, output n integers: the i-th integer denotes the value of $partition(X)$ after the i-th operation.

Example

standard input	standard output
1 10 1 1 1 1 1 2 1 3 1 4 1 5 1 6 1 7 1 8 -2 5	1 1 2 2 3 3 4 4 5 6

Note

The value of X after each operation in the sample: $1, 2, 4, 7, 12, 20, 33, 54, 88, 72$.

Problem 53. Anti-hash Test

Input file:	*standard input*
Output file:	*standard output*
Time limit:	1 second
Memory limit:	256 mebibytes

It is well-known that the following string $s(n) = s_0 s_1 \dots s_{2^n-1}$ can challenge almost every solution that uses polynomial hashes modulo 2^{64}:

$$s_i = \begin{cases} \text{“a”}, & \text{popcount}(i) \bmod 2 = 0 \\ \text{“b”}, & \text{popcount}(i) \bmod 2 = 1 \end{cases}$$

where popcount(i) means the number of ones in binary representation of number i.

Given a string u and an integer n, find the number of occurrences of u in string $s(n)$ and the number of distinct strings v which have the same number of occurrences in string $s(n)$. As both the numbers may be very large, you are only asked to calculate them modulo $10^9 + 7$.

Input

There are multiple test cases. The first line of input contains an integer T, indicating the number of test cases. For each test case:

The first line contains an integer n ($1 \le n \le 10^{18}$).

The second line contains a string u ($1 \le |u| \le \min(10^6, 2^n)$) consisting only of letters “a” and “b”.

It is guaranteed that the sum of $|u|$ over all test cases does not exceed 10^6.

Output

For each test case, if the string u does not appear in string $s(n)$, you should simply output -1. Otherwise, output two integers denoting the the number of occurrences of u in string $s(n)$ modulo $10^9 + 7$ and the number of distinct strings v which have the same number of occurrences in string $s(n)$ modulo $10^9 + 7$.

Example

standard input	standard output
4 10 a 10 abba 5 abbabaabbaababbabaababbaabbabaab 20 ababab	512 2 171 4 1 344 -1

Problem 59. Gardening Game

Input file:	*standard input*
Output file:	*standard output*
Time limit:	15 seconds
Memory limit:	1024 mebibytes

Whiteking is going to play a local decorating game. The area he wants to decorate is in the form of an $N \times N$ grid in which the unit grid is represented by 1×1 square tiles, the coordinates of the top left tile are $(1, 1)$, and the coordinates of the bottom right tile are (N, N). Thus the coordinates of the tile refer to the coordinates of the lower right corner of the tile. From this, it can be seen that the tile located at (x, y) occupies a square area corresponding to $[x - 1, x] \times [y - 1, y]$. Each tile has its own beauty value, initially all tiles have a beauty value of 0.

The local decorating game is a game where the player earns points using the following three actions.

- Divide the grid into different zones by drawing a straight line horizontally or vertically. Initially, there is only one area of size $N \times N$. The lines are infinite in both directions: for example, if you draw a straight line horizontally and another one vertically, the initial area will be divided into a total of four areas.
- Select a tile and decorate the area it belongs to. As a result, the beauty of all tiles in the decorated area is increased by a given value.
- Select a rectangle on the grid and earn points equal to the beauty of the most beautiful tile in it.

Whiteking wants to know in advance how many points he will earn for every action of the third type. So he will show you an ordered list of actions he is going to perform. The actions will be given in the following format:

- "`1` a b": If a is 0, draw the straight line $x = b$, and if a is 1, draw $y = b$.
- "`2` a b X": Select the tile located at (a, b) decorate the area it belongs to, increasing the beauty of all tiles in it by X.
- "`3` a b c d": Select a rectangle with (a, b) as the top left tile and (c, d) as the bottom right tile, and earn points equal to the beauty of the most beautiful tile in it.

Help Whiteking find what will be the score for every action of the third type.

Input

The first line contains two integers N and Q ($1 \le N \le 10^5$, $1 \le Q \le 3 \cdot 10^5$).

Each of the next Q lines describes an action in the format shown in the problem statement.

For an action of type `1`, $0 \le a \le 1$ and $1 \le b \le N - 1$.

For an action of type `2`, $1 \le a, b \le N$ and $-10^9 \le X \le 10^9$.

For an action of type `3`, $1 \le a \le c \le N$ and $1 \le b \le d \le N$.

There are at most 25 000 actions of type 2, and at least 1 action of type 3.

Output

For every action of the third type, print the score that Whiteking will get for that action.

Example

standard input	standard output
3 7 3 1 1 3 3 2 1 3 -3 3 1 1 3 3 1 0 1 2 1 1 4 3 2 2 3 3 3 1 1 3 3	0 -3 -3 1

Problem 61. Rikka with Lake

Input file:	*standard input*
Output file:	*standard output*
Time limit:	5 seconds
Memory limit:	512 mebibytes

Thanks to Peanut and LGD, Rikka won a lot of money during S10. Rikka wants to use the money to buy a piece of land and builds a manor on it.

The map can be regarded as a two-dimensional Cartesian coordinate system. Rikka decides to build her bedroom on point $(0, 0)$. There is a lake on the map, of which the shape is a simple polygon. Of course, point $(0, 0)$ is outside the lake.

Rikka loves running. On each morning, starting from her bedroom, Rikka would firstly run away for k unit of distance and then run back. Therefore, Rikka wishes her manor could be large enough so that no matter how Rikka plans her path, she will always run inside her manor. Note that Rikka will only run on the land since she cannot swim.

Rikka is not a wasteful girl: Rikka wants her manor to be as small as possible. Therefore, she wants you to calculate the minimum possible area of her manor.

Input

The first line contains two integers n, k ($3 \le n \le 50, 1 \le k \le 3000$), representing the number of vertices of the lake and the distance Rikka can run.

Then n lines follow. Each line contains two integers x_i, y_i ($|x_i|, |y_i| \le 200$), representing the coordinates of vertices on the polygon. These vertices are given in clockwise order.

The input guarantees that point $(0, 0)$ is not strictly inside the polygon, and no three continuous vertices of the polygon are on the same line.

Output

Output a single line with a single real number, the minimum possible area of Rikka's manor.

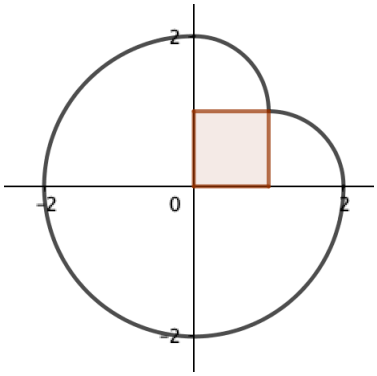
Your answer should have an absolute error or relative error of less than 10^{-6}.

Examples

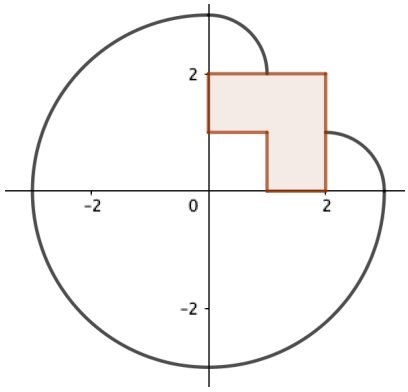
standard input	standard output
4 2 0 0 0 1 1 1 1 0	10.995574287564
6 3 1 0 1 1 0 1 0 2 2 2 2 0	23.776546738526

Note

For the first sample, the minimum possible area of Rikka's minor is $\frac{7}{2}\pi$. One possible plan is shown in the following figure.



For the second sample, the minimum possible area of Rikka's minor is $\frac{29}{4}\pi+1$. One possible plan is shown in the following figure.



Problem 67. Hit the Hay

Input file:	`standard input`
Output file:	`standard output`
Time limit:	5 seconds
Memory limit:	512 megabytes

Some consider putting a baby to sleep to be an art, but this problem will show it is all just maths.

Consider a night during which a parent is trying to put a baby to sleep. An alarm clock will sound at the end of the night, and the parent is not allowed to change the alarm time, so the length of the night is fixed at k hours from now.

The baby can be in one of the three states: state 0 means the baby is awake, state 1 means the baby is in light sleep, and state 2 means the baby is in deep sleep. The baby starts in state 0, and the change in a state happens continuously rather than discretely. You are given three numbers p_0, p_1, and p_2. Whenever the baby is in state i, the probability that no state change will happen in the next x hours is p_i^x, where x is a positive real number. In other words, the time before the next state change is picked from the *exponential distribution* with the cumulative distribution function of $1 - p_i^x$.

Whenever a state change does happen, if the baby was in state 0, it will always switch to state 1; if the baby was in state 2, it will also always switch to state 1; if the baby was in state 1, it will switch to state 0 with the probability q_0 and to state 2 with the probability $1 - q_0$.

The parent decides when to go to sleep themselves. However, if the baby is in state 0, it will cry and wake the parent up, so the parent can only be asleep if the baby is in state 1 or 2. The parent can choose to still stay awake even if the baby is in one of those states. If they do stay awake, they can:

- see which of the three states the baby is in;
- prevent the baby from waking up: if the baby decides to switch from state 1 to state 0 according to the above rules, and the parent is not asleep, then the baby will be comforted and will stay in state 1 instead.

The parent can decide to go to sleep arbitrarily, for example using the current state of the baby or the current time to make this decision. However, if they do go to sleep, then they will be asleep until either the baby wakes up (goes to state 0), or the alarm clock sounds at the end of the k hours. If they get woken up by the baby waking up, then they can later decide to go to sleep again arbitrarily.

What is the maximum expected number of hours of sleep the parent can get if they decide to go to sleep in the optimal fashion?

Input

The first line of the input contains an integer t ($1 \le t \le 1000$) — the number of test cases.

The next t lines describe test cases, each contains five floating-point numbers with exactly *one* digit after the decimal point, in the following order: k, p_0, p_1, p_2, q_0 ($0.1 \le k \le 10$; $0.1 \le p_0, p_1, p_2, q_0 \le 0.9$).

Output

Output t lines with a floating-point number on each line — the maximum expected amount of sleep for each test case. Your outputs will be considered correct if they are within 10^{-9} absolute difference from the answers.

Example

standard input	standard output
2 10.0 0.5 0.5 0.5 0.5 8.0 0.1 0.9 0.9 0.1	6.5990202123649855 7.540407031059442